Stress Tensor, Navier-Poisson & Wall Shear Stress

Cauchy stress tensor decomposition, Newtonian constitutive relations, stress vector projection $f_n = \bar{\bar{\tau}} \cdot n$, and friction drag integration.

OFFICIAL MATERIAL Notes.pdf · p. 20

1. Decomposition of the Cauchy Stress Tensor

In a fluid medium at rest (or moving as a rigid body without deformation), shear stresses are strictly zero and normal stresses are isotropic, characterized solely by the thermodynamic pressure p. The stress tensor is purely spherical:

$$\bar{\bar{\tau}}_{\text{static}} = -p\bar{\bar{I}}$$

[Eq. 3.27] Isotropic hydrostatic stress tensor.

When the fluid undergoes continuous deformation ($\bar{\bar{T}}_d \neq 0$), molecular momentum diffusion gives rise to additional viscous stresses. The total Cauchy stress tensor decomposes universally into:

$$\bar{\bar{\tau}} = -\mathbf{p}\bar{\bar{\mathbf{I}}} + \bar{\bar{\tau}}'$$

[Eq. 3.28] Decomposition into thermodynamic pressure p and viscous deviatoric stress tensor $\bar{\bar{\tau}}'$.

By Cauchy's Stress Principle, the contact force per unit area (stress vector f_n) acting on an imaginary or physical surface element with outward unit normal n is:

$$\mathbf{f_n} = \bar{\bar{\tau}} \cdot \mathbf{n} = -\mathbf{pn} + \bar{\bar{\tau}}' \cdot \mathbf{n}$$

[Eq. 3.19] Stress vector on a surface with normal n.

- **Normal stress:** $\sigma_n = f_n \cdot n = -p + n \cdot \bar{\bar{\tau}}' \cdot n$
- **Tangential (shear) stress vector:** $\tau_s = f_n - \sigma_n n = \bar{\bar{\tau}}' \cdot n - (n \cdot \bar{\bar{\tau}}' \cdot n)n$

Stress Tensor: Complete Derivation Chain (Notes.pdf, Eqs. 3.15–3.25)

Step 1: Geometry of the Cauchy tetrahedron (Fig. 3.2)

The tetrahedron has a vertex at the origin, three faces on the coordinate planes and one oblique face. Its edges along the axes are dx_1, dx_2, dx_3, so the coordinate face normal to e_i has area dA_i and the volume is $dV = \frac{1}{6} dx_1 dx_2 dx_3$ (Notes, p. 28). Let n be the outward unit normal of the oblique face, of area dA. The outward normals of the coordinate faces are $-e_i$. For any closed surface $\oint n \, d\sigma = 0$ (Gauss' theorem applied to a constant field). For the tetrahedron:

$$dA\, n - dA_1 e_1 - dA_2 e_2 - dA_3 e_3 = 0$$

Taking the scalar product with e_i gives the geometrical identity used in the Notes:

$$dA_i = n_i \, dA$$

The volume of a pyramid is one third of base times height, so $dV = \frac{1}{3} h \, dA$, with h the distance from the origin to the oblique face. If all edges are scaled by ε, the face areas scale as ε^2 and the volume as ε^3.

Step 2: Stress vector on opposite faces

Extension, not in the Notes. The Notes state the sign convention directly (the force through the face normal to e_i is $-dA_i f_i$). It follows from Newton's law applied to a thin disc of area $d\sigma$ and thickness ε: the mass and the volume forces scale as $\varepsilon \, d\sigma$ and vanish faster than the two face forces, which gives $f_n \, d\sigma + f_{-n} \, d\sigma \to 0$, that is $f_{-n} = -f_n$. The stress vector on the face with outward normal $-e_i$ is therefore $-f_i$, with $f_i = (\tau_{i1}, \tau_{i2}, \tau_{i3})$.

Step 3: Newton's second law on the tetrahedron and the limit (Eqs. 3.16 and 3.17)

Mass times acceleration equals the surface forces plus the volume force:

$$\rho \, dV \, \frac{Dv}{Dt} = f_n \, dA - f_1 \, dA_1 - f_2 \, dA_2 - f_3 \, dA_3 + \rho f_m \, dV$$

Substitute $dA_i = n_i \, dA$, divide by dA and use $dV/dA = h/3$:

$$f_n - n_1 f_1 - n_2 f_2 - n_3 f_3 = \frac{h}{3} \, \rho \left(\frac{Dv}{Dt} - f_m \right)$$

Let the tetrahedron shrink to a point, $h \to 0$, with the acceleration and the force per unit mass bounded. The right-hand side tends to zero because inertia and volume forces are of order ε^3, whereas the surface forces are of order ε^2. This is Eq. 3.16 divided by dA, and it yields

$$f_n = n_1 f_1 + n_2 f_2 + n_3 f_3$$

Eq. 3.17

In components, $(f_n)_j = \sum_i n_i \tau_{ij}$, so $f_n = n \cdot \bar{\bar{\tau}}$ with $\bar{\bar{\tau}} = [\tau_{ij}]$ (Eq. 3.18). The dependence on n is linear. The symmetry shown in the next step turns it into $\bar{\bar{\tau}} \cdot n$ (Eq. 3.19).

Step 4: Symmetry of the stress tensor (Eq. 3.19)

Apply the angular momentum balance about the centre of a cube of edge dx. The inertia term (moment of inertia of order dx^5) and the moment of the volume force (order dx^4) are negligible against the moment of the surface forces (order dx^3). Consider the z component. The face $x_1 = +dx/2$ has outward normal $+e_1$ and area dx^2; its force has the component $\tau_{12}\, dx^2\, e_2$ at the position $(dx/2)e_1$. The opposite face has force $-\tau_{12}\, dx^2\, e_2$ at $-(dx/2)e_1$. Both give the same moment, using $e_1 \wedge e_2 = e_3$:

$$2 \times \frac{dx}{2}\, e_1 \wedge \left(\tau_{12}\, dx^2\, e_2\right) = \tau_{12}\, dx^3\, e_3$$

The faces normal to e_2 give, with $e_2 \wedge e_1 = -e_3$, the moment $-\tau_{21}\, dx^3\, e_3$. The remaining stress components produce forces parallel to the position vector, or moments perpendicular to e_3. The balance of moments along e_3 is therefore

$$\left(\tau_{12} - \tau_{21}\right) dx^3 = O(dx^4) \implies \tau_{12} = \tau_{21}$$

after dividing by dx^3 and letting $dx \to 0$. The same argument along e_1 and e_2 gives $\tau_{23} = \tau_{32}$ and $\tau_{13} = \tau_{31}$. The tensor has only six independent components and $n \cdot \bar{\bar{\tau}} = \bar{\bar{\tau}} \cdot n$, which is Eq. 3.19.

Step 5: Principal stresses (Eqs. 3.20 and 3.21)

The shear stress on a surface of normal n vanishes if and only if $\bar{\bar{\tau}} \cdot n$ is parallel to n, that is, if

$$\bar{\bar{\tau}} \cdot n = \lambda n \iff \left(\bar{\bar{\tau}} - \lambda \bar{\bar{I}}\right) \cdot n = 0$$

Eq. 3.20

A non-trivial solution exists only if the determinant of the matrix vanishes (Eq. 3.21), a cubic equation in λ. **Extension, not in the Notes.** The roots are real because $\bar{\bar{\tau}}$ is symmetric: for an eigenpair, $\lambda\,\bar{n}\cdot n = \bar{n}\cdot\bar{\bar{\tau}}\cdot n$ is real (its complex conjugate equals itself by symmetry), and $\bar{n}\cdot n > 0$. For two eigenpairs with different eigenvalues,

$$\lambda_1\, n_1\cdot n_2 = (\bar{\bar{\tau}}\cdot n_1)\cdot n_2 = n_1\cdot(\bar{\bar{\tau}}\cdot n_2) = \lambda_2\, n_1\cdot n_2 \implies (\lambda_1-\lambda_2)\, n_1\cdot n_2 = 0$$

so the principal directions are mutually orthogonal.

Step 6: Resultant surface force, Gauss' theorem and the local momentum equation (Eqs. 3.22 to 3.25)

The surface force on a fluid volume V of boundary Σ is the integral of the stress vector (Eq. 3.22). Apply Gauss' theorem (Eq. 2.9) to the vector field with components τ_{ij} ($j=1,2,3$) for each fixed i:

$$\int_\Sigma (\bar{\bar{\tau}}\cdot n)_i\, d\sigma = \int_\Sigma \tau_{ij} n_j\, d\sigma = \int_V \frac{\partial \tau_{ij}}{\partial x_j}\, dV = \int_V (\nabla\cdot\bar{\bar{\tau}})_i\, dV$$

Eq. 3.23

For a volume that shrinks around a point, $\nabla\cdot\bar{\bar{\tau}} = \lim_{V\to 0}\frac{1}{V}\int_\Sigma \bar{\bar{\tau}}\cdot n\, d\sigma$, so $\nabla\cdot\bar{\bar{\tau}}$ is the surface force per unit volume. Newton's second law for a particle of mass $\rho\, dV$ then gives Eq. 3.24, and dividing by dV gives Eq. 3.25:

$$\rho\, dV\,\frac{Dv}{Dt} = (\nabla\cdot\bar{\bar{\tau}})\, dV + \rho f_m\, dV \implies \rho\frac{Dv}{Dt} = \nabla\cdot\bar{\bar{\tau}} + \rho f_m$$

Eqs. 3.24 and 3.25

2. The Navier-Poisson Constitutive Relations

For an isotropic Newtonian fluid (including air and water under standard conditions), viscous stress is linearly proportional to the rate-of-strain tensor $\bar{\bar{T}}_d$:

$$\bar{\bar{T}}_d = \frac{1}{2}\left[\nabla v + (\nabla v)^T\right] \qquad \text{with } T_{d,ij} = \frac{1}{2}\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right)$$

The general isotropic linear tensor constitutive equation is the **Navier-Poisson Equation**:

$$\bar{\bar{\tau}}' = 2\mu\bar{\bar{T}}_d + \lambda(\nabla \cdot v)\bar{\bar{I}}$$

[Eq. 3.30] Navier-Poisson relation with dynamic shear viscosity μ and second viscosity λ.

Expressing λ in terms of the bulk (volume) viscosity $\mu_B = \lambda + \frac{2}{3}\mu$ yields the **Canonical Navier-Poisson Equation**:

$$\bar{\bar{\tau}}' = \mathbf{2}\mu\bar{\bar{\mathbf{T}}}_{\mathbf{d}} + \left(\mu_{\mathbf{B}} - \frac{\mathbf{2}}{\mathbf{3}}\mu\right)(\boldsymbol{\nabla} \cdot \mathbf{v})\bar{\bar{\mathbf{I}}}$$

[Eq. 3.31] Canonical Navier-Poisson equation.

> KEY AERODYNAMIC SIMPLIFICATIONS
>
> **1. Incompressible Flow ($\nabla \cdot v = 0$):** The volumetric dilatation term vanishes identically: $\bar{\bar{\tau}}' = 2\mu\bar{\bar{T}}_d$.
>
> **2. Stokes Hypothesis ($\mu_B \approx 0$):** For dilute monatomic gases, internal molecular relaxation is negligible ($\mu_B = 0 \implies \lambda = -\frac{2}{3}\mu$).

Navier-Poisson Relation: Derivation of Eqs. 3.29-3.31

Step 1: Starting point and principal axes (Eq. 3.29)

A Newtonian fluid has a viscous stress linear in the rate of strain, $\tau'_{ij} = \alpha_{ijkl}\gamma_{kl}$, with $\gamma_{kl} = (\bar{\bar{T}}_d)_{kl}$. The velocity v and the rotation tensor $\bar{\bar{T}}_r$ cannot appear, because a reference frame can always be chosen in which both vanish at a point, whereas the existence of strain rates does not depend on the frame (Notes, p. 31). The rate-of-strain tensor is real and symmetric, so it has three real principal values $\gamma_1, \gamma_2, \gamma_3$ along orthogonal principal axes. For an isotropic fluid the principal axes of $\bar{\bar{\tau}}'$ coincide with those of $\bar{\bar{T}}_d$; call $\tau'_1, \tau'_2, \tau'_3$ the viscous stresses along them.

Step 2: Relation between principal values

Isotropy implies that τ'_1 depends on γ_1 through a coefficient α, and on the two perpendicular strain rates γ_2, γ_3 through one and the same coefficient λ, because no direction is privileged among the two perpendicular ones. Adding and subtracting $\lambda\gamma_1$:

$$\tau'_1 = \alpha\gamma_1 + \lambda\gamma_2 + \lambda\gamma_3 = (\alpha - \lambda)\gamma_1 + \lambda(\gamma_1 + \gamma_2 + \gamma_3)$$

The sum of the principal values is the trace of $\bar{\bar{T}}_d$, which is invariant under rotations and equals $\partial v_i/\partial x_i = \nabla \cdot v$. Defining $\mu = (\alpha - \lambda)/2$ (Notes, p. 31) and repeating the argument for τ'_2 and τ'_3:

$$\tau_i' = 2\mu\,\gamma_i + \lambda\,\nabla\cdot v, \qquad i = 1,2,3$$

Step 3: Tensor form (Eqs. 3.30 and 3.31)

In the principal frame $\bar{\bar{T}}_d$ and $\bar{\bar{I}}$ are diagonal, so the three relations above are the diagonal form of

$$\bar{\bar{\tau}}' = 2\mu\,\bar{\bar{T}}_d + \lambda\,(\nabla\cdot v)\,\bar{\bar{I}}$$

Eq. 3.30

Both sides are tensors, so an identity valid in one orthogonal frame is valid in every orthogonal frame, and the relation holds in general Cartesian coordinates. Introducing the bulk viscosity $\mu_B = \lambda + \frac{2}{3}\mu$, that is $\lambda = \mu_B - \frac{2}{3}\mu$:

$$\bar{\bar{\tau}}' = 2\mu\,\bar{\bar{T}}_d + \left(\mu_B - \frac{2}{3}\mu\right)(\nabla\cdot v)\,\bar{\bar{I}}$$

Eq. 3.31

Step 4: Checks (extension, not in the Notes)

Taking the trace, with $\mathrm{tr}\,\bar{\bar{T}}_d = \nabla\cdot v$ and $\mathrm{tr}\,\bar{\bar{I}} = 3$:

$$\mathrm{tr}\,\bar{\bar{\tau}}' = (2\mu + 3\lambda)\,\nabla\cdot v = 3\mu_B\,\nabla\cdot v$$

so μ_B is the coefficient that links the mean viscous normal stress, $\frac{1}{3}\mathrm{tr}\,\bar{\bar{\tau}}'$, to the dilatation rate. For an incompressible fluid ($\nabla\cdot v = 0$) the relation reduces to $\bar{\bar{\tau}}' = 2\mu\bar{\bar{T}}_d$. Dimensional check: $[\mu] = \mathrm{Pa\,s}$ and $[T_d] = \mathrm{s}^{-1}$, so $[\tau'] = \mathrm{Pa}$.

3. Wall Shear Stress τ_w & Skin-Friction Drag

Consider a viscous boundary layer flowing in the x-direction over a solid flat wall located at $y = 0$ with outward unit normal into the fluid $n = \hat{j}$.

By the no-slip condition, fluid adheres to the wall: $v_x(x,0) = 0$ and $v_y(x,0) = 0$. Along the solid wall, $\frac{\partial v_y}{\partial x} \equiv 0$. The tangential viscous shear stress exerted on the wall simplifies to:

$$\tau_{\mathbf{w}} = \mu \left. \frac{\partial \mathbf{v_x}}{\partial \mathbf{y}} \right|_{\mathbf{y=0}}$$

The total **viscous skin-friction drag** D_f acting on a wetted surface of length L and span W is obtained by integrating the local wall shear stress:

$$\mathbf{D_f} = \int_{\mathbf{\Sigma}} \tau_{\mathbf{w}} \, \mathbf{d}\sigma = \mathbf{W} \int_{\mathbf{0}}^{\mathbf{L}} \tau_{\mathbf{w}}(\mathbf{x}) \, \mathbf{dx}$$

In aerospace aerodynamics, this is parameterized by the skin friction drag coefficient C_f:

$$C_f = \frac{D_f}{\frac{1}{2}\rho v_\infty^2 (WL)}$$

Wall Shear Stress from the Stress Tensor (Notes.pdf, Eqs. 3.17, 3.28, 3.30 and 3.32)

Extension, not in the Notes (worked application of Eqs. 3.30 and 3.32). Consider a flat wall at $y = 0$ with the fluid at $y > 0$. The unit normal pointing into the fluid is $n = e_y$. By Eq. 3.32 the force per unit area exerted by the fluid on the wall is the stress vector

$$f = \bar{\bar{\tau}} \cdot e_y = -p\, e_y + \bar{\bar{\tau}}' \cdot e_y$$

In components, Eq. 3.30 reads $\tau'_{ij} = \mu \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right) + \lambda (\nabla \cdot v)\, \delta_{ij}$, since $(T_d)_{ij} = \frac{1}{2}(\partial v_i/\partial x_j + \partial v_j/\partial x_i)$. The tangential component of the force is

$$f_x = \tau'_{xy} = \mu \left(\frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x} \right) + \lambda (\nabla \cdot v)\, \delta_{xy}$$

The Kronecker delta is zero for $x \neq y$, so the bulk-viscosity term does not contribute to shear. The no-slip condition gives $v_y(x, 0) = 0$ for every x, hence its derivative along the wall vanishes, $\partial v_y / \partial x|_{y=0} = 0$. Therefore

$$\tau_w = f_x = \mu \left. \frac{\partial v_x}{\partial y} \right|_{y=0}$$

The normal component is $f_y = -p + \tau'_{yy}$ with $\tau'_{yy} = 2\mu\, \partial v_y/\partial y + \lambda \nabla \cdot v$. At the wall $v_x(x, 0) = 0$ for every x, so $\partial v_x/\partial x = 0$ and $\nabla \cdot v = \partial v_y/\partial y$. For an incompressible fluid $\nabla \cdot v = 0$, hence $\partial v_y/\partial y = 0$, $\tau'_{yy} = 0$ and $f_y = -p$: only pressure loads the wall in the normal direction, and the viscous contribution is purely tangential. The friction drag is the integral of τ_w over the wetted surface (second term of Eq. 3.32 projected on the flow direction).

CANONICAL PROBLEM WALKTHROUGH

Braking Torque in a Rotating Couette Viscometer

Physical Setup: A precision Couette (or Searle) rotational viscometer consists of two coaxial concentric cylinders of length L. The inner cylinder has radius R_1 and rotates at steady angular velocity Ω. The outer cylinder has radius R_2 and is fixed. The annular gap between cylinders $h = R_2 - R_1$ is very narrow ($h \ll R_1$), filled with a liquid of unknown viscosity μ.

Objective: Derive the viscous braking torque $\mathcal{M}$ required to maintain steady rotation, and express dynamic viscosity μ in terms of measurable laboratory quantities.

1 Velocity Gradient across the Narrow Annular Gap

Because the gap is narrow ($h/R_1 \ll 1$), curvature effects can be neglected locally. The flow in the gap is an unrolled plane Couette flow with azimuthal velocity $v_\theta(r)$:

- At the inner rotating cylinder ($r = R_1$): $v_\theta = \Omega R_1$.
- At the outer stationary cylinder ($r = R_2 = R_1 + h$): $v_\theta = 0$.

The velocity varies linearly across the gap $y = r - R_1$:

$$v_\theta(r) = \Omega R_1 \left(1 - \frac{r - R_1}{h}\right)$$

The transverse shear rate is uniform:

$$\left| \frac{dv_\theta}{dr} \right| = \frac{\Omega R_1}{h}$$

2 Wall Shear Stress on the Inner Cylinder

Applying Newton's law of viscosity at the inner cylinder surface $r = R_1$:

$$\tau_w = \mu \left| \frac{dv_\theta}{dr} \right| = \mu \frac{\Omega R_1}{h}$$

This shear stress acts uniformly over the entire cylindrical wetted surface area $A = 2\pi R_1 L$, opposing rotation.

3 Integrate Viscous Braking Torque $\mathcal{M}$

Each surface element $dA = R_1 d\theta dz$ experiences a friction force $dF = \tau_w dA$ with a lever arm $r = R_1$ from the rotation axis. Integrating over the surface:

$$\mathcal{M} = \int_A R_1 \tau_w \, dA = R_1 \tau_w (2\pi R_1 L) = 2\pi R_1^2 L \tau_w$$

Substituting $\tau_w = \mu \frac{\Omega R_1}{h}$ yields the analytical **Couette Viscometer Torque Formula**:

$$\mathcal{M} = \frac{\mathbf{2\pi \mu \Omega R_1^3 L}}{\mathbf{h}}$$

4 Inversion for Laboratory Viscosity Measurement

In the laboratory, measuring the applied torque $\mathcal{M}$ via a torsion wire or load cell and the angular velocity Ω directly determines the dynamic viscosity:

$$\mu = \frac{\mathcal{M}\mathbf{h}}{\mathbf{2\pi \Omega R_1^3 L}}$$

NOTE: THIN-GAP APPROXIMATION AND THE EXACT RESULT

The result $\tau_w = \mu\Omega R_1/h$ is the **thin-gap approximation**; its relative error is $O(h/R_1)$. In cylindrical coordinates the exact shear stress is $\tau_{r\theta} = \mu\, r\, \dfrac{d}{dr}\left(\dfrac{v_\theta}{r}\right)$ (rigid rotation, $v_\theta = \omega r$, gives zero stress). The azimuthal momentum balance for steady circular flow gives $v_\theta = Ar + B/r$. The conditions $v_\theta(R_1) = \Omega R_1$ and $v_\theta(R_2) = 0$ give $B = \Omega R_1^2 R_2^2/(R_2^2 - R_1^2)$. Since $v_\theta/r = A + B/r^2$, then $\tau_{r\theta} = -2\mu B/r^2$. At $r = R_1$, $|\tau_{r\theta}| = 2\mu\Omega R_2^2/(R_2^2 - R_1^2)$, and the torque ($r^2\tau_{r\theta}$ is independent of r) is

$$\mathcal{M}_{\text{exact}} = \frac{4\pi\mu\Omega L\, R_1^2 R_2^2}{R_2^2 - R_1^2}, \qquad \frac{\mathcal{M}_{\text{exact}}}{\mathcal{M}_{\text{thin}}} = \frac{2(R_1+h)^2}{R_1(2R_1+h)} \approx 1 + \frac{3h}{2R_1}$$

With $R_2^2 - R_1^2 = h(2R_1 + h) \to 2R_1 h$ for $h \ll R_1$ the exact torque reduces to the thin-gap formula above (extension, verified symbolically).

AEROSPACE EXAM INSIGHT: MECHANICAL DISSIPATION POWER

The mechanical power consumed by the motor to overcome viscous friction is:

$$\dot{W}_{\text{diss}} = \Omega\mathcal{M} = \frac{2\pi\mu\Omega^2 R_1^3 L}{h}$$

By the First Law of Thermodynamics (Subchapter 3.5), this mechanical power is completely converted into heat by viscous dissipation inside the fluid gap, causing the lubricant's temperature to rise if cooling is not provided!

← 3.5 Total Energy | Topic 3 Hub | Back to Hub & Cheat Sheet →

Study notes written from the official course material (Notes.pdf · p. 20). Online version: subjects/fluid-mechanics/teoria/topic-3-6-stresses.html